

Owned Requirements: FR-001–FR-008, SEC-001–SEC-005

Owned Components: Auth/User Module, Complaint Module

Specialist Section: 5.1 Security Validation

5.1 Security Validation

Security validation will verify that authentication, authorization, input validation, session protection and complaint attachment controls satisfy the security requirements defined in the SRS.

SEC-001 – Password Protection

Verify that passwords are not stored in plaintext and that stored credentials use the configured secure password representation.

SEC-002 – Role-Based Access Control

Verify that residents cannot access staff or administrator functionality and that protected operations enforce role permissions.

SEC-003 – Input Validation

Submit malformed, unexpected or invalid input through registration and complaint forms and verify that the system validates and safely handles the input.

SEC-004 – Session Security

Attempt to access protected pages or APIs using invalid, expired or missing authentication sessions and verify that access is rejected.

SEC-005 – Attachment Security

Attempt to upload supported and unsupported file types and files exceeding the configured size restriction. Verify that only permitted attachments are accepted.

Security testing will include both valid and invalid scenarios. High-priority defects affecting authentication, authorization or protected data access should be treated as release-blocking issues.

13. Test Case Management & Traceability

13.1 Test Cases – Auth/User Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-AUTH-01	FR-001	Register with valid details	Registration page available	Enter all required valid details and submit	Resident account is created successfully	Positive
TC-AUTH-02	FR-001	Register with already registered account details	Registration page available	Enter an email address that is already registered and submit the form	Account creation is rejected with appropriate validation message	Negative

TC-AUTH-03	FR-002	Registration with missing required field is rejected	Registration page available	Leave one required field empty and submit the form	Registration is rejected and an appropriate validation message is displayed	Positive
TC-AUTH-04	FR-002	Registration with invalid information is rejected	Registration page available	Enter invalid information, such as an incorrectly formatted email address, and submit the form	Registration is rejected and an appropriate validation message is displayed	Negative
TC-AUTH-05	FR-003	Login with valid credentials	Registered resident exists	Enter valid email/password and login	Authenticated session is established	Positive
TC-AUTH-06	FR-003	Login with incorrect password	Registered resident exists	Enter valid email and incorrect password	Login is rejected	Negative
TC-AUTH-07	FR-004	Access permitted resident functionality	Resident successfully authenticated	Access resident dashboard/complaint functionality	Permitted functionality is accessible	Positive
TC-AUTH-08	FR-004	Resident attempts staff/admin function	Resident successfully authenticated	Attempt to access protected staff/admin function	Access is denied	Negative

13.2 Test Cases – Complaint Module

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-CMP-01	FR-005	Submit valid complaint	Resident authenticated	Enter valid complaint details and submit	Complaint is successfully created	Positive
TC-CMP-02	FR-005	Unauthenticated user attempts complaint submission	User not authenticated	Attempt to submit complaint	Submission is rejected and authentication is required	Negative
TC-CMP-03	FR-006	Submit complaint with all mandatory fields	Resident authenticated	Enter title, category, description and location	Complaint is accepted	Positive
TC-CMP-04	FR-006	Submit complaint with missing mandatory field	Resident authenticated	Leave one mandatory field empty and submit	Submission is rejected with validation message	Negative
TC-CMP-05	FR-007	Upload permitted attachment	Resident authenticated	Attach a supported file within configured size limit	Attachment is accepted	Positive
TC-CMP-06	FR-007	Upload unsupported/oversized attachment	Resident authenticated	Attach unsupported or oversized file	Attachment is rejected	Negative

TC-CMP-07	FR-008	Generate Complaint ID after successful submission	Resident authenticated	Submit valid complaint	System generates a unique Complaint ID	Positive
TC-CMP-08	FR-008	Verify unique Complaint IDs are generated	Existing complaints available	Submit multiple valid complaints	Each successfully submitted complaint receives a different Complaint ID	Positive

13.3 Test Cases – Security

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type
TC-SEC-01	SEC-001	Verify secure password storage	Registered user exists	Inspect stored credential representation	Password is not stored in plaintext	Positive
TC-SEC-02	SEC-001	Verify plaintext password is not stored in the database	Registered user exists	Inspect the database record for the registered user's password	The password is stored only in a secure hashed representation and is not readable as plaintext	Negative
TC-SEC-03	SEC-002	Resident accesses permitted functionality	Resident authenticated	Access resident functions	Resident functionality is allowed	Positive
TC-SEC-04	SEC-002	Resident attempts privileged functionality	Resident authenticated	Attempt staff/admin operation	Access is denied	Negative
TC-SEC-05	SEC-003	Submit valid input	Registration/complaint form available	Enter valid expected input	Input is accepted and processed	Positive
TC-SEC-06	SEC-003	Submit malformed/unexpected input	Registration/complaint form available	Enter malformed or unsafe input	Input is rejected or safely handled	Negative
TC-SEC-07	SEC-004	Access protected page with valid session	User authenticated	Access protected functionality	Request is accepted	Positive
TC-SEC-08	SEC-004	Access protected page with expired/invalid session	User session expired/invalid	Attempt protected request	Request is rejected	Negative
TC-SEC-09	SEC-005	Upload permitted attachment	Resident authenticated	Upload supported file within configured size	Attachment is accepted	Positive
TC-SEC-10	SEC-005	Upload prohibited attachment	Resident authenticated	Upload unsupported/oversized file	Attachment is rejected	Negative

13.4 Requirement Traceability

Requirement ID	Component	Test Case IDs
FR-001	Auth/User Module	TC-AUTH-01, TC-AUTH-02
FR-002	Auth/User Module	TC-AUTH-03, TC-AUTH-04
FR-003	Auth/User Module	TC-AUTH-05, TC-AUTH-06
FR-004	Auth/User Module	TC-AUTH-07, TC-AUTH-08
FR-005	Complaint Module	TC-CMP-01, TC-CMP-02
FR-006	Complaint Module	TC-CMP-03, TC-CMP-04
FR-007	Complaint Module	TC-CMP-05, TC-CMP-06
FR-008	Complaint Module	TC-CMP-07, TC-CMP-08
SEC-001	Auth/User Module	TC-SEC-01, TC-SEC-02
SEC-002	Auth/User Module	TC-SEC-03, TC-SEC-04
SEC-003	Auth/User + Complaint Module	TC-SEC-05, TC-SEC-06
SEC-004	Auth/User Module	TC-SEC-07, TC-SEC-08
SEC-005	Complaint Module	TC-SEC-09, TC-SEC-10

13.5 Test Data / Evidence to Collect

- Registration screenshots and automated test output for valid and invalid registration.
- Authentication evidence for successful and failed login attempts.
- Authorization evidence showing resident denial for staff/administrator functions.
- Evidence that passwords are not stored in plaintext.
- Input validation evidence for malformed registration and complaint data.
- Session-security evidence for invalid or expired sessions.
- Attachment validation evidence for permitted, unsupported and oversized files.
- Evidence of unique Complaint IDs for successfully submitted complaints.